B.M.S. COLLEGE OF ENGINEERING

(Autonomous Institute, Affiliated to VTU)

Bull Temple Road, Basavanagudi, Bengaluru - 560019

An AAT Project Report on n

"Online Complaint Management System"

Submitted in partial fulfilment of the requirements for the award of degree

BACHELOR OF ENGINEERING

IN

INFORMATION SCIENCE AND ENGINEERING

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Under the guidance of

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Assistant Professor

Department of Information Science and Engineering
2025-2026

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Department of Information Science and Engineering

C E R T I F I C A T E

This is to certify that the project entitled "Online Complaint Management System" is a bona-fide work carried out by Chiranth C (1BM23IS071), Devendra E (1BM23IS079) and Gagan Chandra R (1BM23IS090) in partial fulfilment of the requirements for the award of Bachelor of Engineering in Information Science and Engineering from Visvesvaraya Technological University during the academic year 2024–2025

Rashmi K B	**Dr. M K Nalini**	**Dr. Bheemsha Arya**
Assistant Professor	**Professor and HOD**	**Principal**

Examiners

Name of the Examiner	**Signature of the Examiner**
1.	
2.	

DECLARATION

We, Chiranth C (1BM23IS071), Devendra E (1BM23IS079) and Gagan Chandra R (1BM23IS090), students of B.E. Information Science and Engineering, B.M.S. College of Engineering, Bengaluru, hereby declare that the project entitled "Online Complaint Management System" is an authentic work carried out under the guidance of Rashmi K B, Assistant Professor, Department of Information Science and Engineering.

The work presented in this report has not been submitted to any other university or institution for the award of any degree or diploma

Place: Bengaluru
Date:06-01-2025

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ABSTRACT

The Online Complaint Management System (OCMS) is a secure web-based platform developed to allow citizens to register complaints digitally and track case status in real time. The system supports multiple departments such as Financial Fraud, Cyber Crime, Women and Child Protection, Kidnapping and Missing Person cases. Users can submit detailed complaints with supporting evidence, while authorized officers can manage and update case status through a dedicated dashboard. The project is implemented using Flask, SQLite, HTML/CSS and ReportLab.

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CHAPTER 1: INTRODUCTION

In today's digital era, governance systems are rapidly shifting towards online platforms to improve efficiency, transparency and accessibility. Crime reporting is one such critical service that traditionally requires victims to physically visit police stations. This approach often leads to delays, lack of proper tracking and inconvenience to victims. The Online Complaint Management System bridges this gap by providing a centralized online portal for complaint registration and monitoring.

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CHAPTER 2: SYSTEM DESIGN

The system follows a classic web based client–server architecture. The main components are:

• User Interface: Implemented using HTML, CSS and Jinja2 templates rendered by Flask. It provides pages for home, login, registration, complaint submission and dashboards. • Application Layer: Developed using Flask. It handles routing, form processing, session management, validation and PDF generation. • Database Layer: Implemented using SQLite with tables for users, complaints and OTP codes. It stores user credentials, complaint metadata, evidence file names, phone numbers and timestamps.

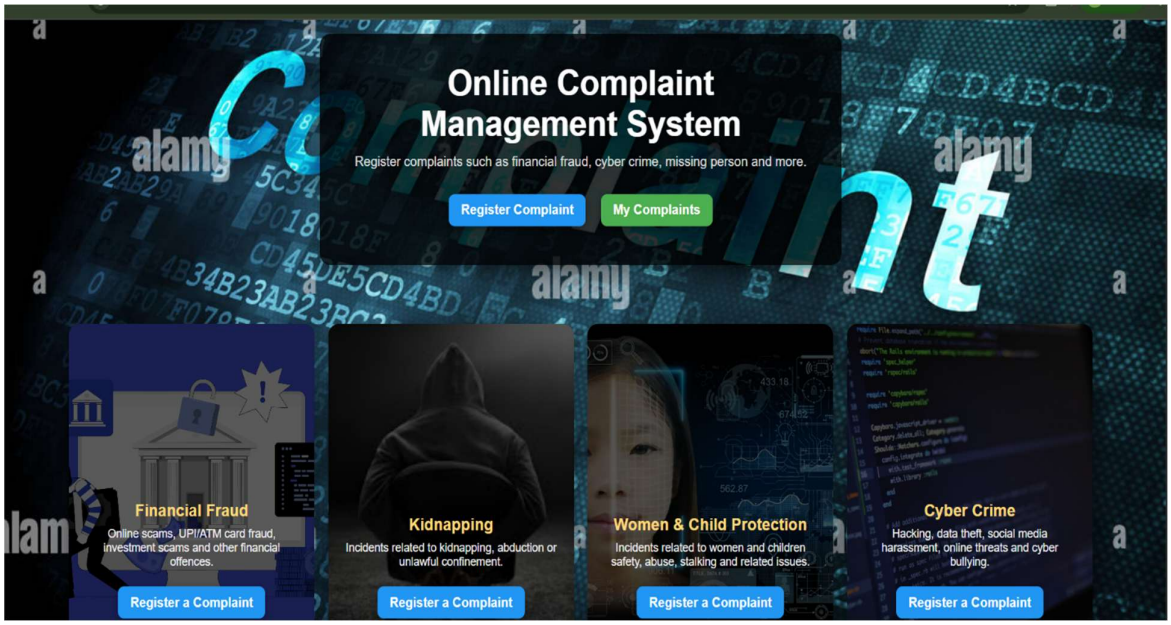
The complaints table stores fields such as department, incident date and time, location, victim details, accused details, description, evidence file name, phone number, status and timestamps for creation and last update. Helper modules like user_db.py and complaint_db.py encapsulate all database operations.

CHAPTER 3 : IMPLEMENTATION

3.1 FRONT END:

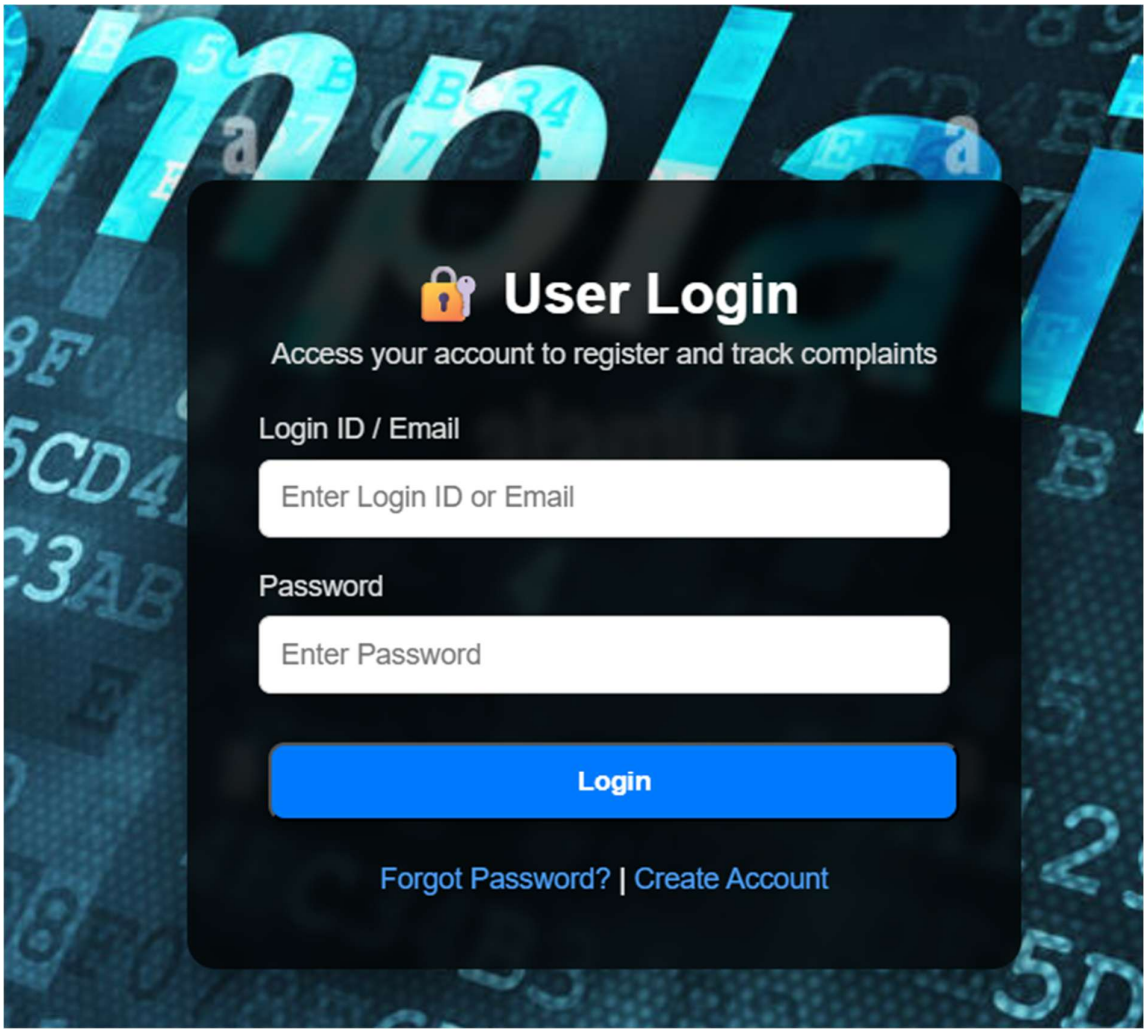
HOME PAGE:

The home page highlights the title "Online Complaint Management System" and provides quick access buttons to register a new complaint or view existing complaints. It also displays cards for different departments such as Financial Fraud, Kidnapping, Women & Child Protection and Cyber Crime to guide the user.



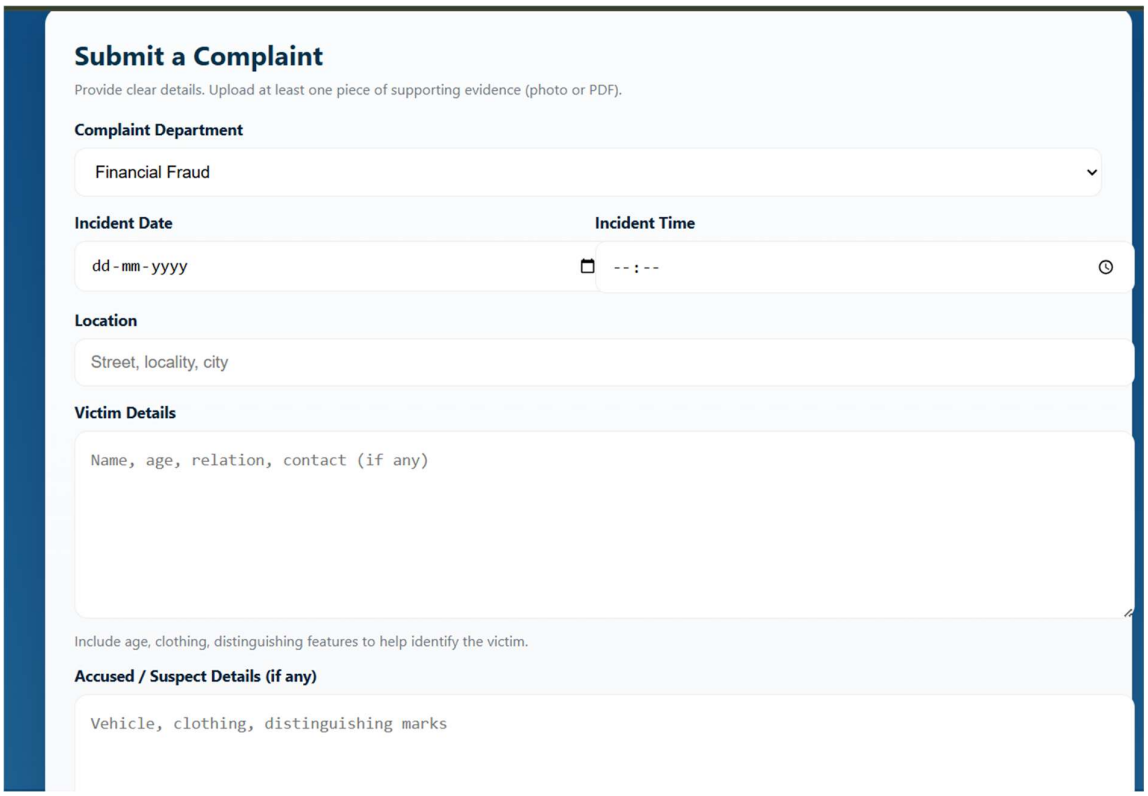
User Login Screen

The user login screen allows registered users to securely access the system using their login ID or email and password. Links are provided for forgotten password and account creation, making the system accessible for new as well as existing users.

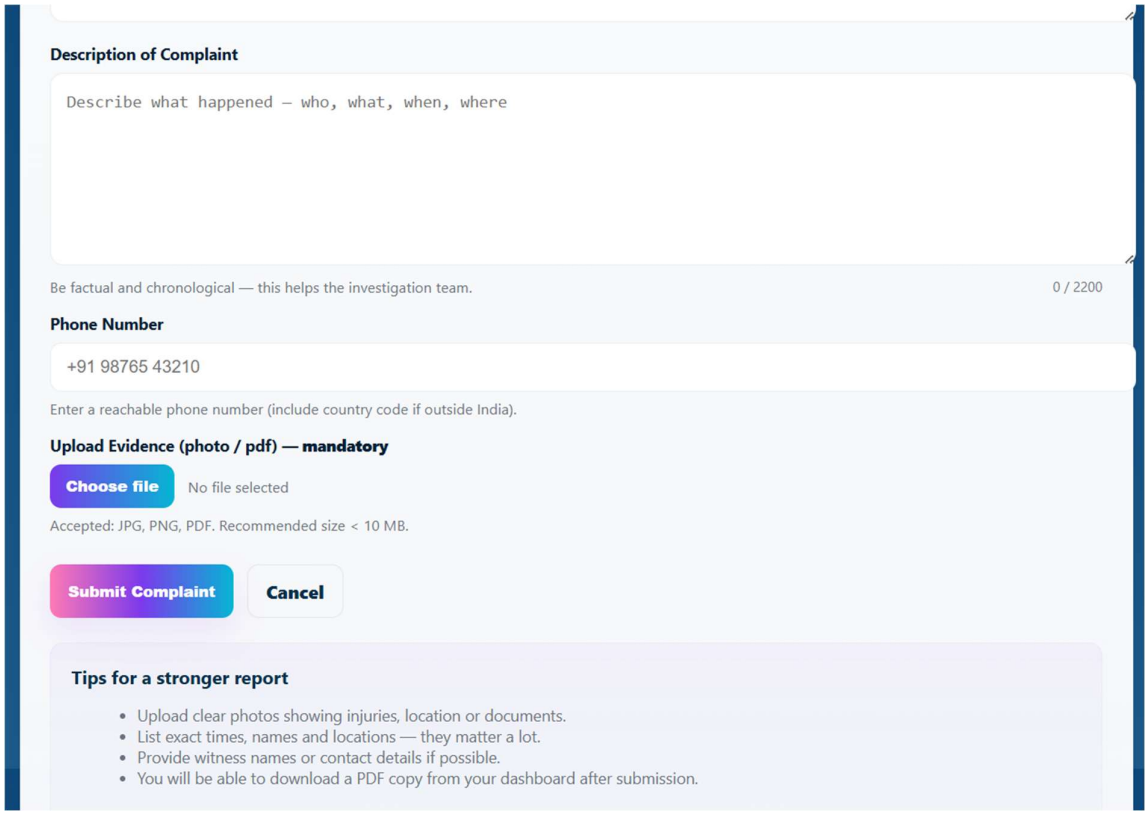


Submit Complaint – Form View

The complaint submission form collects key information such as department, incident date and time, location, victim details and accused details. This structured data helps the investigation team understand the context of the incident clearly

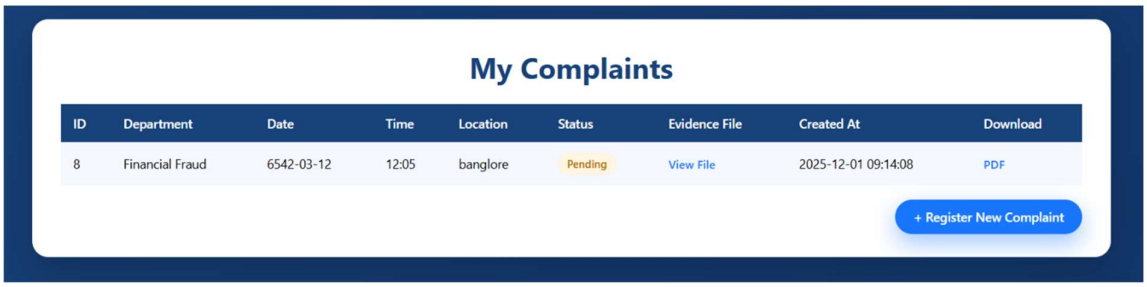


The lower part of the form captures a detailed description of the complaint, the reporter's phone number and the mandatory evidence upload field which accepts images or PDF files. Guidelines are provided to help users submit stronger, more useful reports.



My Complaints – User Dashboard

After submitting complaints, users can visit the "My Complaints" page to see a tabular view of all their cases. For each complaint, the table shows department, date, time, location, status, evidence file link, creation timestamp and a link to download the complaint as PDF.



My Complaints

ID	Department	Date	Time	Location	Status	Evidence File	Created At	Download
8	Financial Fraud	6542-03-12	12:05	banglore	Pending	View File	2025-12-01 09:14:08	PDF

+ Register New Complaint

Department Case Handling – Officer Dashboard

Department officers access a dedicated dashboard which lists all complaints belonging to their department. The table shows reporter username, name, phone, department, incident date and time, evidence file and current status. Officers can update status using a drop down menu and also download files for detailed review.

Department Case Handling

Department: Financial Fraud

#	Username	Reporter / Name	Phone	Department	Date	Time	File	Status	Action
1	shivu	wdw	6345124578	Financial Fraud	6542-03-12	12:05	1764540848_Complaint_Management_System_Report (1).pdf	Pending	Update Download
2	Deva	chiranth	8073804948	Financial Fraud	2024-10-10	10:10	1764240255_complaint_6.pdf	Resolved	Update Download
3	rohit	lwjq	8073804948	Financial Fraud	2024-12-10	10:10	1764205955_kidnap.jpg	Resolved	Update Download
4	rohit	lwjq	8073804948	Financial Fraud	2024-12-10	10:10	1764205930_kidnap.jpg	Resolved	Update Download
5	rohit	Chiranth C	8073804948	Financial Fraud	2024-12-10	11:00	1764164089_kidnap.jpg	Resolved	Update Download
6	rohit	chra	8073804948	Financial Fraud	2025-02-25	10:10	1764159101_kidnap.jpg	Resolved	Update Download
7	rohit	drdrdrdr	-	Financial Fraud	2005-09-08	12:00	-	In Progress	Update Download
8	rohit	rohit	-	Financial Fraud	2025-03-03	10:10	-	In Progress	Update Download

3.2 Back End Overview

The backend is implemented in Python using Flask. The main routes are defined in main.py. Separate helper modules user_db.py and complaint_db.py keep database logic modular and reusable. File uploads are stored inside an uploads directory with timestamp based file names to avoid collisions. The complaints table is created and maintained by complaint_db.ensure_schema(). Functions like add_complaint(), view_complaints_by_user(), view_complaints_by_department() and update_complaint_status() implement all CRUD operations required by the portal.

CHAPTER 4

PYTHON TECHNIQUES USED

• Flask Framework – used for routing, template rendering, handling form submissions, session management and flash messages. • SQLite3 – used as an embedded relational database for storing users, complaints and OTP related information. • ReportLab – used to generate downloadable PDF reports for individual complaints. • datetime and os modules – used for timestamping records and safely managing file paths for uploaded evidence. • JSON responses officer status updates can be performed via an AJAX friendly JSON API.

CHAPTER 5

CONCLUSION

The Online Complaint Management System successfully digitises the end to end workflow of crime complaint registration and tracking. Citizens can submit complaints conveniently, attach digital evidence and monitor status without repeated physical visits to a police station. Officers benefit from a structured dashboard that groups cases by department, supports quick status updates and provides ready to download PDF summaries. The project demonstrates how modern web technologies such as Flask, SQLite and responsive web design can be combined to build practical e governance solutions. The system can be extended in future with SMS/email notifications, advanced analytics and integration with external law enforcement databases.

CHAPTER 6: REFERENCES

- Flask Documentation – https://flask.palletsprojects.com/
- SQLite Documentation – https://sqlite.org/docs.html
- ReportLab User Guide – https://www.reportlab.com/documentation/
- GeeksforGeeks Flask Tutorials – https://www.geeksforgeeks.org/flask-tutorial/